

Lab 2, task #3 (Monday 04.05.2026)

Using the pymoo framework (in Python):

- 1) implement solutions for the following Single-Objective Optimization (SOO) tasks using a genetic algorithm (GA, population size 100 elements):

- a) minimization of the **Three-hump Camel Function** (requires implementation of your own class to handle the problem) – **max. 2 points**:

$$F(x_1, x_2) = 2x_1^2 - 1.05x_1^4 + \frac{x_1^6}{6} + x_1x_2 + x_2^2, x_i \in [-5; 5]$$

where the global minimum is achieved for decision variables $x_i^* = 0$ with the objective function value $F^*(x_1^*, x_2^*) = 0$;

- b) minimization of **Sphere function** (available directly in pymoo) – **max. 1 point**:

where the global minimum is achieved for $x_i^* = 0$ with the objective function value $F^*(x_1^*, x_2^*) = 0$;

$$F(x_1, x_2) = \sum_{i=1}^2 x_i^2, x_i \in [-10; 10]$$

- 2) repeat the procedure for solving tasks 1a) and 1b) at least 20 times for

- 100 generations,
- 1,000 generations,
- 5,000 generations,

calculate the average values and standard deviations (for each number of generations separately) of the obtained objective function value F^* – **max. 3 points**;

- 3) write a report on the above work containing – (**max. 2 points**) including:

- fitness landscape graphs (type “surface”) of the functions studied in 1a) and 1b);
- results of the research from point 2;
- commentary on the results obtained.